In Catilinam IV ⋆

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Abstract

Cum M. Cicero consul Nonis Decembribus senatum in aede Iovis Statoris consuleret, quid de iis coniurationis Catilinae sociis fieri placeret, qui in custodiam traditi essent, factum est, ut duae potissimum sententiae proponerentur, una D. Silani consulis designati, qui morte multandos illos censebat, altera C. Caesaris, qui illos publicatis bonis per municipia Italiae distribuendos ac vinculis sempiternis tenendos existimabat.

Key words: Cicero; Catiline; orations.

1 Introduction

Gauge invariance is a central organizing principle of modern physics, expressing the idea that observable quantities should remain unchanged under certain local transformations of the mathematical description. In classical field theories and quantum electrodynamics, this concept ensures that different representations related by gauge transformations correspond to the same physical reality [3,5]. From an abstract viewpoint, gauge invariance provides a systematic way to separate physically meaningful dynamics from redundant coordinate or potential offsets. This symmetry-based perspective has proven essential for constructing robust and consistent models in physics, and it motivates the extension of invariance principles beyond their traditional domain toward engineered dynamical systems subject to measurement distortions and unknown offsets.

2 Preliminaries

2.1 The System Model

Let $x \in \mathbb{R}^{n\times 1}$, $y \in \mathbb{R}^{p\times 1}$ and $w \in \mathbb{R}^{r\times 1}$ be the state, measurement, exogenous vectors, respectively, the SIMO/MIMO LTI system addressed in this paper is stated as,

$$\dot{x} = Ax + B_w w,\ y = Cx + D_w w \tag{1}$$

where $A \in \mathbb{R}^{n\times n}$, $B_w \in \mathbb{R}^{n\times r}$, $C \in \mathbb{R}^{p\times n}$ and $D_w \in \mathbb{R}^{p\times r}$. The following additional rank condition

$$\mathrm{rank}(C) < p \tag{2}$$

arises in sparse sensor applications[4], topologies used in Multi Agen Systems(MAS)[2] and distributed networks [7], and is also called Strictly Output Redundant(SOR) system [6].

2.2 The Luenberger Observer

The Luenberger Observer is defined as,

$$\dot{\hat{x}} = A\hat{x} + L(y - \hat{y}),\ \hat{y} = C\hat{x} \tag{3}$$

where $L \in \mathbb{R}^{n\times p}$ is the observer gain. Assuming (A, C)-pair observable, the error dynamics are obtained for the

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error $e \triangleq x - \hat{x}$ as,

$$\dot{e} = (A - LC)e + (B_w - LD_w)w \tag{4}$$

The $\mathcal{H}_\infty$ optimal observer for objective function $z = C_z e$ is designed using the following LMI problem[1],

$$\min_{Y,P} (\gamma) \quad \text{s.t.}$$

$$\begin{bmatrix} (PA - YC) + \star & PB_w - YD_w & C_z^T \\ \star & -\gamma I & 0 \\ \star & 0 & -\gamma I \end{bmatrix} \prec 0$$

$$P \succ 0$$

where the observer gain is recovered with $L = P^{-1}Y$.

3 Projection Filtering

3.1 The Projection

The projection operator defined using Gauge-Invariance principle is "proposed" as follows,

$$\Pi^C \triangleq I - C(C^TC)^{-1}C^T \tag{5}$$

which satisfies,

$$\Pi^C C = (I - C(C^TC)^{-1}C^T)C = 0 \tag{6}$$

Let $C^\perp = \mathcal{N}(C^T)$,

$$\Pi^C C^\perp = (I - C(C^TC)^{-1}C^T)C^\perp = C^\perp \tag{7}$$

and for the projection the following holds,

$$(\Pi^C)^2 = \Pi^C,\ \Pi^C \in \mathbb{R}^{p\times p} \tag{8}$$

Using the projection operator on the measured output yields,

$$\Pi^C y = \Pi^C(Cx + D_w w) = \Pi^C D_w w \tag{9}$$

The projection eliminates the state dependent component of the output, isolating the exogenous component establishing state invariance. This is an algebraic measurement of the disturbance reflected to the output.The exogenous input is recovered under the following condition,

$$\mathcal{R}(D_w) \not\subseteq \mathcal{R}(C) \tag{10}$$

so that

$$\Pi^C D_w \neq 0 \tag{11}$$

with the least square estimator,

$$\hat{\omega} = (\Pi^C D_w)^\dagger(\Pi^C y) \tag{12}$$

where $(.)^\dagger$ is the Moore-Penrose pseudo-inverse. This recovery is possible if and only if,

$$\mathcal{N}(\Pi^C D_w) = \{0\} \tag{13}$$

4 Numerical Example

4.1 Example 1

The electrical system given in[6] with parameters $C_t = 2.2$mF, $L_t = 1.8$mH and $R_t = 0.2\Omega$ is defined as the following state space model,

$$A = \begin{bmatrix} 0 & \frac{1}{C_t} \\ -\frac{1}{L_t} & -\frac{-R_t}{L_t} \end{bmatrix},\ B_w = \begin{bmatrix} -\frac{1}{C_t} \\ 0 \end{bmatrix},\ C = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{14}$$

where output redundancy is created by replicating the first output sensor as the second output sensor. The exogenous input matrix is chosen by combining the orthogonal component of output matrix with the output matrix as,

$$D_w = C^\perp + C\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.7071 \\ 0.2929 \\ 1.0000 \end{bmatrix} \tag{15}$$

A Luenberger observer is designed solving the LMI given in Eq 2.2 and is obtained as,

$$L = \begin{bmatrix} -326.1332 & 366.9908 & -5.2903 \\ -33.4833 & 28.1920 & 48.9023 \end{bmatrix} \tag{16}$$

noting that $B_w - LD_w = 0$ is satisfied, hence the state estimation is not affected by the exogenous input. The estimated state can be used to obtain the exogenous input by utilizing,

$$\hat{w} = D_w^\dagger(y - Cx) \tag{17}$$

For the projection filtering the projection matrix given by Eq 5 is calculated for C as,

$$\Pi^C = \begin{bmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{18}$$

where the projection for D_w is,

$$\Pi_w^D = \begin{bmatrix} 0.2714 & -0.1250 & -0.4268 \\ -0.1250 & 0.9786 & -0.0732 \\ -0.4268 & -0.0732 & 0.7500 \end{bmatrix} \tag{19}$$

The state is filtered using,

$$\hat{x} = (\Pi_w^D C)^\dagger (\Pi_w^D y) \tag{20}$$

and the exogenous input is filtered using,

$$\hat{w} = (\Pi^C D_w)^\dagger (\Pi^C y) \tag{21}$$

The state estimation and the projection state filtering is compared in Figure 1.

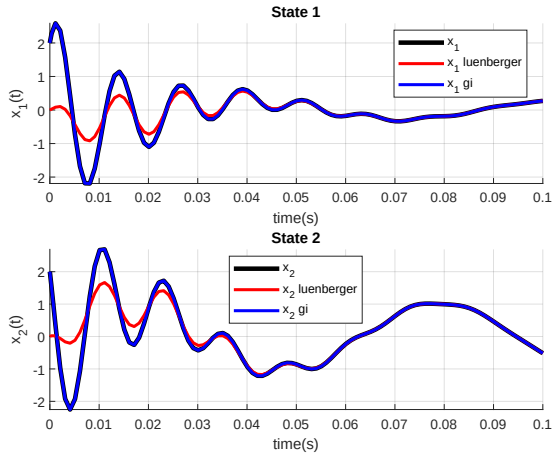

Fig. 1. State filtering versus estimation of Example 1.

The exogenous input estimated and filtered is depicted in Figure 2.

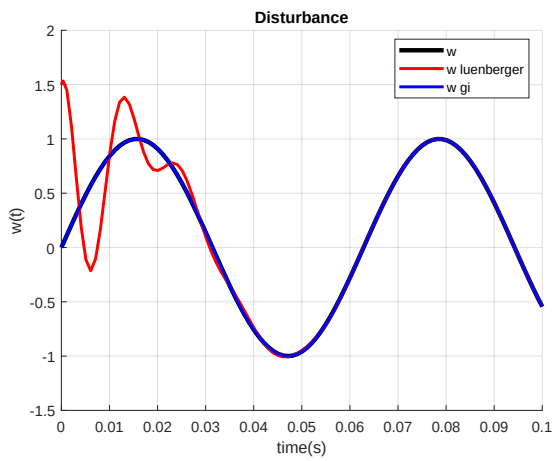

Fig. 2. Disturbance

The projection filtering method utilizes the geometric properties of the output mapping, hence no dynamic estimation is present, it directly extracts the output map into its components. Despite the fact that the Luenberger observer gain for output redundant systems is able to eliminate the exogenous input term in the error dynamic by setting $B_w - LD_w$ to zero, the structure exploit performs better for the given example since no estimation takes place. Static output filtering fails if

$$D_w = C\alpha \tag{22}$$

since,

$$\Pi y = \Pi C x + \Pi D_w w = \Pi C(x + \alpha) = 0 \tag{23}$$

which shows that different channels need to exist in order separate signals using the projection.

4.2 Example 2

Let the output mapping,

$$y = Cx + D_w w + D_n n \tag{24}$$

be given as,

$$y = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 2.2060 \\ 2.3970 \\ 1.3015 \\ 1.3015 \end{bmatrix} w(t) + \begin{bmatrix} 0.1960 \\ 0.7382 \\ 0.6697 \\ 0.5265 \end{bmatrix} n(t) \tag{25}$$

where $w(t)$ is disturbance, $n(t)$ is noise and $x(t)$, which represents the system states, for illustrative purposes is chosen as,

$$x(t) = \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} \tag{26}$$

The output mapping is modified as,

$$y = Cx + \begin{bmatrix} D_w & D_n \end{bmatrix} \begin{bmatrix} w \\ n \end{bmatrix} \tag{27}$$

and the projection is chosen as,

$$\Pi^{wn} = I - \begin{bmatrix} D_w & D_n \end{bmatrix} \left(\begin{bmatrix} D_w \\ D_n \end{bmatrix} \begin{bmatrix} D_w & D_n \end{bmatrix} \right) \begin{bmatrix} D_w \\ D_n \end{bmatrix} \tag{28}$$

$$\Pi^{wn} = \begin{bmatrix} 0.6224 & -0.2581 & -0.3034 & -0.2763 \\ -0.2581 & 0.1118 & 0.1698 & 0.0618 \\ -0.3034 & 0.1698 & 0.5564 & -0.3549 \\ -0.2763 & 0.0618 & -0.3549 & 0.7094 \end{bmatrix} \tag{29}$$

The state $x(t)$ is recovered using,

$$\hat{x} = (\Pi^{wn} C)^{\dagger} (\Pi^{wn} y) \tag{30}$$

The output filtering results are depicted in Figure 3.

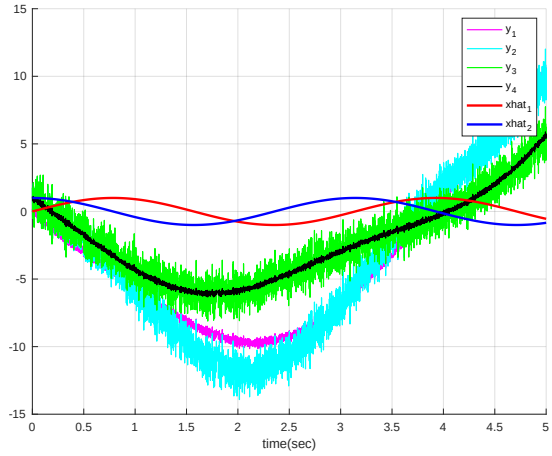

Fig. 3. State recovery

The disturbance is recovered using the projection,

$$\Pi^{xn} = I - \begin{bmatrix} C & D_n \end{bmatrix} \left(\begin{bmatrix} C \\ D_n \end{bmatrix} \begin{bmatrix} C & D_n \end{bmatrix} \right) \begin{bmatrix} C \\ D_n \end{bmatrix} \tag{31}$$

$$\Pi^{xn} = \begin{bmatrix} 0.7254 & -0.3627 & 0.1507 & 0.2120 \\ -0.3627 & 0.1813 & -0.0754 & -0.1060 \\ 0.1507 & -0.0754 & 0.0313 & 0.0440 \\ 0.2120 & -0.1060 & 0.0440 & 0.0619 \end{bmatrix} \tag{32}$$

as depicted in Figure 4. The state and disturbance re-

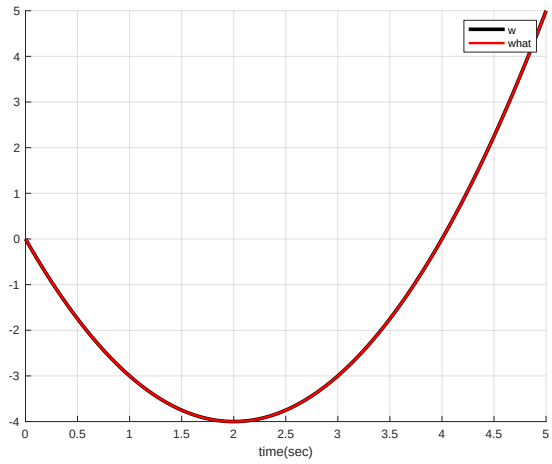

Fig. 4. Disturbance recovery

covery is successful as it can be observed from Figure 3 and Figure 4.

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A A summary of Latin grammar

B Some Latin vocabulary